

CareConnect Software Requirements Specification (SRS)

OMNICHANNEL CROSS-PLATFORM ENGINEERING & ARCHITECTURAL SPECIFICATION

1. Project Framing & Design Philosophy

The **CareConnect** platform is a cross-platform care coordination application designed for patients and caregivers living with severe Carpal Tunnel Syndrome (CTS) and peripheral nerve compression pathologies. Unlike standard health-tech systems that optimize for age-related cognitive decline, CareConnect is built upon a fundamental architectural axiom: **the user's cognition is fully intact, but their hands are the bottleneck**. Users think clearly and remember details perfectly; however, they cannot type, tap repeatedly, or hold hardware devices without severe median nerve pain.

This specification establishes a strict, non-segregated framework across all deployment platforms—Mobile (Flutter/React Native), Desktop (Electron/React), and Responsive Web (React/PWA). Every platform variant is built for all three primary user roles: patients, family caregivers, and professional medical staff, removing any arbitrary feature barriers across different form factors.

***Universal System Constraint:** Pain is the primary usability adversary. Every interaction model across mobile, desktop, and web must minimize mechanical friction and nerve compression from the first code commit.*

2. Enforced Core Architectural Constraints

To pass binary compliance review, every view and layout template must strictly adhere to the following four system constraints across all repositories:

- **C1. Voice-First Input:** Text fields, system searches, and core navigation modules must feature complete voice alternatives. Users can execute commands or input notes purely by speaking, making voice a native, first-class interface layout element.
- **C2. Large, Accessible Touch Targets:** All buttons and inputs must maintain a strict physical target minimum size of 60x60 pixels / 88dp, accompanied by clear safe margin paddings. Interactive targets are clustered in ergonomic high-access zones to eliminate awkward wrist stretching.
- **C3. No Sustained or Repetitive Gestures:** Interactive events requiring actions longer than 200ms are completely barred from the codebase. Long-press behaviors are replaced by unambiguous two-tap confirmation loops. Drag-and-drop handles and infinite scrolling list patterns are replaced by manual pagination controls.

- **C4. Hardware-Flexible Operation:** Interfaces must scale flawlessly to accommodate external device arrays, including vertical ergonomic mice, specialized keypads, phone stands, and assistive operating system access engines (such as iOS Voice Control, Android Voice Access, macOS Voice Control, and Windows Voice Access).

3. Comprehensive Omnichannel Personas

All platform runtimes must be engineered to satisfy the overlapping constraints of the following three verified user archetypes:

3.1 Semi Kim (The High-Activity Professional Lens)

- **Context & Background:** 35-year-old public school teacher managing over 100 students daily, logging health metrics under tight schedules between active classes.
- **Physical Constraints:** Continuous wrist pain, index finger numbness, tingling, and hand tremors that degrade touchscreen tap precision.
- **Cross-Platform Expectations:** Expects mobile alerts to be actionable in seconds via knuckle strikes on large targets; requires desktop tracking layouts to prioritize rapid, template-driven keyword selections over text strings.
- **Technical Skill Level:** Has experience using multiple apps and digital tools.

3.2 Eleanor "Ellie" Vance (The High-Cognitive Coordination Lens)

- **Context & Background:** 58-year-old retired retail manager operating as the primary care coordinator for her post-stroke father, managing high administrative stress.
- **Physical Constraints:** Severe bilateral carpal tunnel and early osteoarthritis; acute thumb joint flares and hand fatigue peaking in the late afternoon.
- **Cross-Platform Expectations:** Demands mobile layouts with zero multi-touch gesture requirements; demands desktop environments to display complex schedules, histories, and notes side-by-side to eliminate tedious navigation click loops.
- **Technical Skill Level:** Moderate and practical skill with typing, inventory scanning, and data logging.

3.3 Alex Lewis (The High-Velocity Automated Data Lens)

- **Context & Background:** 35-year-old Information System Security Officer (ISSO) reviewing compliance frameworks, monitoring system metrics, and compiling long data reports.
- **Physical Constraints:** Intense nerve pain and hand numbness brought on by prolonged computer usage and repetitive mouse clicking.
- **Cross-Platform Expectations:** Demands complete, mouse-free workflow systems relying on clear keyboard hotkey layouts; requires voice macros on mobile devices to bypass long navigation scrolling menus.
- **Technical Skill Level:** Advanced technical proficiency; relies heavily on desktop and laptop environments. Experienced with security systems, risk management tools, documentation, and continuous computer-based workflows.

4. Feature Architecture: Adaptive Layout Blueprints

To accommodate the omnichannel strategy, the application layout changes between mobile execution states and wide-screen desktop/web review configurations. Below are the precise adaptive component blueprints for the two primary high-interaction features.

FEATURE A: SMART NOTE-TAKER FRAMEWORK (/APPOINTMENTS)

This module captures fast-moving clinical guidance during medical consultations. Typing and manual text formatting are prohibited.

- **Mobile Application Layout Profile:** Optimized for single-handed, phone-on-stand operation. The interface features a massive, centrally balanced audio recording status radar taking up 60% of the screen. When voice capture is active, the component flashes an ultra-high-contrast pulsing visual ring, providing clear confirmation without requiring touch checks. Starting and stopping are handled via an oversized target button or a native voice command. Text fields are entirely read-only on mobile devices.
- **Desktop & Web Layout Profile:** Implements a responsive, side-by-side split-column dashboard view. The left panel houses real-time wave monitoring utilities, hotkey control lists, and live speech-to-text preview displays. The right panel features a structured markdown parser that automatically breaks down dictated notes into clean, bulleted sub-headers (e.g., *"Medication Changes," "Action Items," "Follow-up Schedules"*), completely preventing large blocks of raw text. All nodes feature single-key copy bindings for mouse-free data transfer.

FEATURE B: HEALTH LOG METRIC LEDGER (/HEALTHLOG)

Tracks daily vitals, symptom spikes, and pain metrics without relying on standard numerical keyboard typing or precise sliders.

- **Mobile Application Layout Profile:** Elements are arranged in a strict single-column layout centered within the natural bottom-screen thumb interaction zone. Traditional numeric entry boxes and sliding pins are replaced by massive, isolated mechanical step-control blocks. A large [+] button sits on the right, and a large [-] button sits on the left, flanking a centralized, large-text read-only indicator. Tapping anywhere near the target increments values safely, avoiding input errors.
- **Desktop & Web Layout Profile:** Arranged in a multi-column view. The left section provides a column of categorized template cards (e.g., *"No Pain," "Mild," "Severe"*), allowing users to quickly select metrics using keyboard numeric keys. The right section displays a wide, high-contrast historical log trend chart. This timeline graph maps data using oversized intersection plots (minimum 12px diameters), allowing users to review health logs side-by-side with schedule history charts without changing views.

FEATURE C: CENTRAL DASHBOARD HUD (/DASHBOARD)

Serves as the primary daily check-in hub, filtering information overload down to immediate daily priorities and actionable targets.

- **Mobile Application Layout Profile:** Stacked in a strict, linear single-column arrangement of oversized quick-action tiles spanning 100% of the viewport width. Secondary informational data fields are actively hidden. The top half of the display is reserved exclusively for a single, high-visibility "Next Due Action" banner card, while the lower half displays a permanent voice command listener bar and a persistent emergency access trigger.
- **Desktop & Web Layout Profile:** Arranged in a multi-column grid matrix layout that surfaces data simultaneously to minimize navigation clicks. The left quad displays a side-by-side view of today's medication tracking rows, the center pane displays the upcoming 48-hour calendar timeline, and the right-hand panel renders a dedicated workspace widget for incoming caregiver messages. A global command dashboard responds instantly to keyboard macros, allowing mouse-free platform management.

FEATURE D: MEDICATIONS MANAGER (/MEDICATIONS)

Tracks prescription adherence, dosages, and administration routines without forcing long-press triggers or precise menu selection.

- **Mobile Application Layout Profile:** Renders a scroll-free, swipe-free card carousel positioned inside the lower thumb access zone. Each medication card features an oversized pill icon and a direct voice-note execution trigger. Adherence tracking discards standard slide-to-confirm or hold-to-confirm paths, relying instead on an explicit two-tap confirmation sequence (Tap 1: *Log Taken* -> Tap 2: *Confirm Status*).
- **Desktop & Web Layout Profile:** Formatted as a dense, high-contrast spreadsheet-style inventory ledger optimized for rapid keyboard sorting. Users navigate rows via the Up/Down arrow keys and toggle compliance logs using the Spacebar. A side panel instantly exposes caregiver voice-note waveforms, audio playback nodes, and complete historical compliance graphs matching the selected line item without shifting active layout contexts.

FEATURE E: EMERGENCY CRISIS HUB (/EMERGENCY)

Provides instantaneous connection to emergency response teams and primary caregivers under physical distress.

- **Mobile Application Layout Profile:** Displays a stark, full-screen, two-element interface. The top half is a single, giant 911 dispatch button measuring 50% of the screen height, and the bottom half is an identical-scale primary caregiver speed-dial card. The screen ignores all swipe interactions; an alert is instantly escalated through a rapid two-tap sequence struck anywhere inside the target zone.
- **Desktop & Web Layout Profile:** Implements a persistent, system-wide escape protocol layout. Hit via a global keyboard shortcut (e.g., `Ctrl + Shift + E`), the app instantly shifts to an ultra-high-visibility emergency interface displaying local geolocated coordinates, primary medical history data blocks, and caregiver contact rosters in massive text strings. It features a large keyboard-focused countdown cancel bar to catch accidental activations.

FEATURE F: CAREGIVER MESSAGES HUB (/MESSAGES)

Facilitates communications between patients, family members, and professional clinical staff without precision typing requirements.

- **Mobile Application Layout Profile:** Formatted as a "face-first" grid layout composed of large avatar contact targets matching the 88dp baseline standard. Tapping a contact card automatically opens an audio-first messaging prompt. Users communicate entirely via voice messaging or voice-to-text dictation, using a simple spoken macro command like *"Call Sarah"* to execute an instant outbound call.
- **Desktop & Web Layout Profile:** Utilizes a standard split-pane master-detail arrangement. The left pane displays a keyboard-navigable index of active care communication channels, while the right pane opens the message thread. Incoming text is automatically paired with an on-screen text-to-speech reader widget. Outbound replies support templated text strings and macro autocomplete selections, allowing complex clinical updates to be dispatched mouse-free in under three keystrokes.

5. Comprehensive Route & Functional Screen Specifications

The core scope of CareConnect consists of eight synchronized functional screens, built with the four core constraints applied to every framework repository from day one:

ROUTE & VIEW	CORE OPERATIONAL REQUIREMENTS	CROSS-PLATFORM ACCESSIBILITY ADAPTATIONS
<code>/</code> Landing Page	Product introduction view displaying core system features, licensing, and an explicit Progressive Web App (PWA) installation panel.	Zero manual text inputs or typing configurations are required to clear promotional pop-ups or navigate layouts.
<code>/signin</code> Authentication	Secure entry portal with clear local form validation tools. Displays accessible pre-filled demo credentials for grading workflows.	Inputs feature voice-fill commands and large target boxes. Login is executed via a single large tap target or hotkey.
<code>/dashboard</code> Central HUD	The primary home center displaying summary action tiles for today's medications, immediate calendar schedules, and a permanent voice listener bar.	Displays immediate tasks directly on launch without initial navigation steps. Tiles span the screen width to maximize touch area.
<code>/medications</code> Med Manager	Full add, track, and delete medication portal. Detail views display clear pill images alongside audio-recorded instructions from caregivers.	Hold-to-confirm slides are replaced with a clear two-tap 'Log-as-Taken' confirmation loop to prevent hand strain.
<code>/emergency</code> Crisis Hub	Dedicated, critical emergency screen featuring a giant 911 dispatch button and primary caregiver speed-dial alert cards.	A rapid two-tap pattern instantly escalates alerts without requiring any long-press or steady grip interactions.
<code>/healthlog</code> Ledger	System tracking ledger for pain fluctuations, vital signs, and symptoms, paired with a cleanly paginated, historical log view.	No infinite scrolling. Replaces sliders with large step targets, utilizing pre-built status checkboxes to minimize typing.
<code>/appointments</code> Calendar	Unified calendar schedule tracking clinical appointments and checkups, filtered cleanly by date matrices and clinician metadata tags.	Integrates the voice-activated Smart Note-Taker framework, allowing family or clinical staff to pin voice alerts directly to the schedule.
<code>/messages</code> Comms Center		

	Communication center featuring a face-first contact matrix and simplified direct voice macro call connections.	Supports simple spoken voice triggers (e.g., " <i>Call Sarah</i> ") to initiate calls without requiring any list scrolling.
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6. Rigid WCAG 2.1 Level AA Accessibility Mapping Plan

Compliance is verified programmatically via automated testing and strict design system guidelines across all four framework repositories:

6.1 Success Criterion 2.1.1 — Keyboard Access Optimization

- **Requirements Parameter:** 100% of functional capabilities must be executable via standard keyboard controls without mouse dependency.
- **Code Enforcement:** Interactive nodes use native button elements. Hidden skip navigation hooks are added at the top level of pages to bypass header lists. Focus rings must be styled with a high-contrast 3px border outline. Focus traps are treated as compilation blockers.

6.2 Success Criterion 2.5.3 & 4.1.2 — Labeling for Speech Infrastructure

- **Requirements Parameter:** All interactive components must feature clear, visible text strings matching their voice activation tags to remain discoverable by native OS voice engines.
- **Code Enforcement:** Standalone graphic icons or abstract symbols are prohibited. Buttons must map their programmatic names to their visible text strings. Forms must include explicit, permanent text label components.

6.3 Success Criterion 2.5.5 — Target Sizing and Spacing

- **Requirements Parameter:** Expand interactive touch areas past standard requirements to account for hand tremors and finger numbness.
- **Code Enforcement:** Every clickable component must measure a minimum size of 60x60 pixels on desktop/web interfaces and 88dp within the natural bottom thumb zone on mobile platforms. Target containers must maintain an 8px safe border buffer to prevent accidental adjacent inputs.

6.4 Success Criterion 1.4.3 & 1.4.4 — Visibility & Text Resizing

- **Requirements Parameter:** System fonts must scale up to 200% without clipping content boundaries, while maintaining strict high-contrast text visibility.
- **Code Enforcement:** Enforce high-contrast text options across all color themes, meeting a minimum ratio of 4.5:1 (WCAG AA). Color cannot be the single indicator used to represent status alterations. Core text defaults must start at a 16pt baseline using relative scaling units to support layout scaling up to 200%.

4.1.2 & 4.1.3 — Screen Reader & Assistive Technology Support

- **Requirements Parameter:** All interactive components must expose a programmatic name, role, and state that screen readers (NVDA on Windows, TalkBack on Android, VoiceOver on iOS/macOS) can announce correctly. Dynamic status changes — a medication logged, a voice command recognised, an emergency alert sent — must be announced automatically via ARIA live regions without requiring the user to navigate to find the update.
- **Code Enforcement — Semantic structure:** Every page uses a single `<h1>` for the page title, semantic landmark elements (`<main>`, `<nav>`, `<header>`), and native HTML elements (`<button>`, `<label>`, `<table>`) wherever possible. DOM reading order must match visual layout — CSS positioning must not create a mismatch. Heading levels must not be skipped; the heading tree is the primary navigation mechanism

for screen reader users. [Maps to: WCAG 1.3.1, 1.3.2, 2.4.6]

- **Code Enforcement — ARIA roles and states:** Custom components use WAI-ARIA only when a native HTML element cannot provide the required semantics. CareConnect-specific requirements: step-controls (/healthlog) use role=spinbutton with aria-valuenow/min/max; voice recording radar (/appointments) uses role=status with aria-label; the 911 button (/emergency) carries aria-label='Call emergency services — 911'; the contact grid (/messages) uses role=grid with role=gridcell per card. All icon-only controls have an aria-label matching the visible tooltip or voice command trigger. All accordion/dropdown toggles use aria-expanded=true/false. [Maps to: WCAG 4.1.2, 2.5.3]
- **Code Enforcement — Live region announcements:** A polite aria-live region is present on every route for non-urgent confirmations (medication logged, note saved, calendar filter applied, voice command recognised). An assertive aria-live region handles time-sensitive announcements that must interrupt current speech: recording started, emergency alert sent, medication already logged today, voice command not recognised. aria-atomic=true is set on all status banners so the full message is re-read on each update. [Maps to: WCAG 4.1.3]
- **Code Enforcement — Focus management:** SPA route transitions programmatically move focus to the new page <h1> so screen reader users are immediately oriented when the view changes. The two-tap confirmation dialogs on /medications and /emergency trap focus within the dialog until the user confirms or cancels, then return focus to the triggering element. Validation errors on /signin move focus to the first invalid field. Voice command completions that reload the view follow the same focus-to-h1 rule. [Maps to: WCAG 2.1.1, 2.4.3]
- **Code Enforcement — Images and charts:** All non-decorative images carry descriptive alt text. Pill images (/medications): alt text identifies the medication (e.g. 'Round white tablet — Aspirin 500mg'). Contact photos (/messages): alt text uses the contact's name and action (e.g. 'Sarah Lewis — tap to call'). The historical trend chart (/healthlog): exposes a text summary sentence or a toggleable accessible data table — a canvas or SVG with no alternative fails WCAG 1.1.1. Purely decorative images use alt="" to be ignored by screen readers. [Maps to: WCAG 1.1.1]
- **Code Enforcement — Flutter and React Native:** Flutter: every custom widget that presents meaningful information or is interactive must be wrapped in a Semantics widget with label, hint, button/adjustable role, and value properties. SemanticsRole enum assigns correct roles (SemanticsRole.button, SemanticsRole.slider) that map to TalkBack and VoiceOver roles on device. React Native: all interactive components set accessibilityLabel (announced name), accessibilityHint (action description, e.g. 'Double tap to log as taken'), accessibilityRole (button, link, adjustable, header), and accessibilityLiveRegion (polite or assertive) on dynamic content containers. [Maps to: WCAG 4.1.2, 4.1.3]
- **Testing:** Automated: axe-core (zero violations required, already in CI pipeline) and eslint-plugin-jsx-a11y added to all React and React Native repos to catch ARIA misuse at write-time. Manual: a ten-item screen reader test protocol is executed on NVDA (Windows), TalkBack (Android), and VoiceOver (iOS/macOS) at Phase 2 Sprint 4 (initial) and Phase 3 Sprint 6 (release sign-off). Protocol covers: route heading announcements, step-control value changes, live region firing on all 8 routes, focus management on dialogs and route transitions, chart/image alt text, and error focus. All ten items must pass on all three screen readers before the release gate is cleared.

7. Technical Architecture, State Management, & Testing Approach

- **Data Handling & Local Enclaves:** CareConnect relies on a zero-backend layout, completely removing remote cloud authentication databases or network request dependencies. Runtime data maps are managed on the client side using local storage arrays and `localStorage` hooks, securing privacy and ensuring immediate access without password screens.

- **Verification Pipeline & Quality Gates:** Automated testing pipelines are integrated directly within the continuous integration build workflows. Every pull request runs automated accessibility checks via `axe-core`, `axe DevTools`, and Google Lighthouse testing engines. Pull requests are automatically blocked if any view drops below the required coverage markers (60% code coverage minimum for Electron/Flutter; 75% code coverage minimum for React Native/Web) or if a Google Lighthouse accessibility score drops below 90. Final validation includes manual testing using native operating system screen readers (NVDA on Windows, TalkBack on Android, and VoiceOver on iOS/macOS).

Deployment Roadmap

This document presents the CareConnect 12-week engineering roadmap, compressed from the original 20-week plan. Three phases of two sprints each (2 weeks per sprint) deliver infrastructure, all 8 product routes, and full WCAG 2.1 AA certification.

Color key

- Green — infrastructure, developer tooling, monorepo, CI
- Blue — feature builds (routes, UI components, voice layer)
- Amber — accessibility, QA, and compliance work
- Red/coral — risks, open questions, and blockers
- Purple gate banners — CI/QA milestones required before the next phase begins

Summary timeline

Phase	Sprints / Weeks	Goal	Gate
Phase 1	S1–S2 Wks 1–4	Infrastructure, CI, voice spike, platform decision	CI green · Lighthouse ≥90 baseline
Phase 2	S3–S4 Wks 5–8	All 8 routes functional; reference feature implementations	All routes render · constraint audits pass
Phase 3	S5–S6 Wks 9–12	WCAG AA certified, QA hardening, release candidate	Screen reader sign-off · Lighthouse ≥90 · RC tagged

Phase 1 — Foundation & Infrastructure

Timeline: Sprints 1–2 | Weeks 1–4

Goal: Full CI pipeline live, monorepo scaffolded, voice spike resolved, platform (Flutter vs RN) decided, design token system published. Zero feature code merged until every gate is green.

Repo / DX

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 1 Wks 1–2	Monorepo + scaffold nx/turborepo, /packages/ui-core Establish monorepo (Nx or Turborepo). Create /packages/ui-core enforcing C1–C4 tokens. Scaffold four repos: Electron+React, React PWA, React Native or Flutter (confirm Sprint 1). Add ESLint rule blocking non-compliant touch	C2, C4	High priority

	targets.		
Sprint 2 Wks 3–4	CI pipeline + tokens axe-core, Lighthouse ≥90, coverage gates Wire axe-core + Lighthouse into every PR (block if < 90). Coverage gates: 60% Electron/Flutter, 75% RN/Web. Publish design tokens: --touch-target-min 60px/88dp, 16pt base font, 4.5:1 contrast minimum. Gesture-audit checklist in PR template.	C2, C3, C4; WCAG 2.1 AA Deps: Monorepo + scaffold	Blocker Lighthouse CI config varies per platform — spike in Sprint 1

Voice layer

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 1 Wks 1–2	Voice abstraction spike Web Speech API + Whisper.js fallback Validate Web Speech API cross-browser coverage. Prototype Whisper.js WASM fallback if Firefox support is absent. Define VoiceCommandRegistry interface mapping spoken phrases to screen actions (Open Q #1 — must close this sprint).	C1	Blocker Web Speech API has no Firefox support — fallback must be resolved Sprint 1
Sprint 2 Wks 3–4	VoiceCommandRegistry + voice bar Natural language mapping + HUD component Implement VoiceCommandRegistry (fuzzy match, default commands: navigate, log, call). Build persistent voice listener bar for Dashboard HUD (idle/listening/processing/error states, 60×60px tap target, OS voice control compatible).	C1, C2, C4 Deps: Voice abstraction spike	Sprint 2

Data + platform

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 1 Wks 1–2	Platform decision + state schema Flutter vs RN benchmark; localStorage design Run performance benchmark on mid-range Android hardware (startup, jank, 88dp accuracy).	C2, C4	Blocker Unresolved — gates all mobile scaffold. Must close Sprint 1 (Open Q #3).

	Decide Flutter vs RN by end of Sprint 1. Simultaneously define client-side state schema: slices per route, serialised to localStorage on each mutation. Multi-device sync is out of scope.		
Sprint 2 Wks 3–4	Mobile scaffold + PWA shell + axe baseline Bottom-thumb zone; service worker; Lighthouse pass Scaffold mobile app in chosen framework with bottom-thumb-zone layout. Implement PWA service worker caching all 8 route shells (offline support). Run axe-core sweep + Lighthouse audit on all 8 routes; fix all violations to establish CI gate baseline.	C2, C3, C4; WCAG 2.1 AA Deps: Platform decision; Voice bar	Milestone

GATE Phase 1 gate: CI green on all repos · Lighthouse ≥90 baseline · voice fallback confirmed · platform decision made

Phase 2 — Core Feature Builds

Timeline: Sprints 3–4 | Weeks 5–8

Goal: All 8 routes functional. Smart Note-Taker and Health Log Ledger built as reference constraint implementations. Every route passes gesture audit before Phase 3.

Smart Note-Taker — /appointments

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 3 Wks 5–6	Audio capture + markdown parser Web Worker, speech-to-text, auto-segment Build recording engine in a Web Worker (start/stop via button ≥60×60px or voice command). Mobile: full-screen radar, 60% viewport, pulsing ring, read-only fields. Implement structured markdown segmentation: Medication Changes, Action Items, Follow-up Schedules.	C1, C2, C3 Deps: VoiceCommandRegistry	Sprint 3 Web Worker + SharedArrayBuffer may need COOP/COEP headers
Sprint 4 Wks 7–8	Single-key copy + voice alert pin Mouse-free output; calendar attach Single-key copy bindings on all structured output nodes. Keyboard shortcuts visible as UI labels (WCAG 2.5.3). Allow family/clinical staff to pin voice alerts to calendar appointments	C1, C3; WCAG 2.5.3, C4 Deps: Audio capture + markdown parser	Sprint 4

	via two-tap loop — no long-press, no mouse required.		
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Health Ledger — /healthlog

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 3 Wks 5–6	Step-controls + template card selector [–] display [+]; keyboard card selection Replace all numeric/slider inputs with large step-control blocks (40px tap halo for tremor tolerance, bottom thumb zone on mobile). Desktop: categorised template cards selectable by keyboard numeric keys. Serialise to same data model.	C2, C3, C4 Deps: Design token system	Sprint 3
Sprint 4 Wks 7–8	Trend chart + screen reader audit Recharts, 12px dots, paginated; TalkBack + VoiceOver Historical trend chart (Recharts/Victory, ≥12px markers, paginated 7/30-day, no infinite scroll). Side-by-side with schedule history. Manual screen reader walkthrough: TalkBack (Android), VoiceOver (iOS). Fix all violations.	C3; WCAG 2.1 AA Deps: Step-controls	Sprint 4

Remaining 6 routes

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 3 Wks 5–6	/signin, /dashboard, /medications, /emergency Demo creds; HUD tiles; two-tap confirm; 911 button /signin: pre-filled demo creds, voice-fill, single-tap login. /dashboard: immediate task tiles, full-width targets, permanent voice bar. /medications: two-tap 'Log-as-Taken' loop, pill images, audio instructions, no long-press. /emergency: giant 911 button, two-tap caregiver speed-dial, no long-press.	C1, C2, C3 Deps: Voice bar; State schema	Sprint 3
Sprint 4 Wks 7–8	/messages, / Landing + PWA install Voice-trigger calls; install panel; RISK: spec incomplete	C1, C2, C3 Deps: Offline / PWA shell	Risk ⚠ /messages spec incomplete — schedule clarification before Sprint 4

	/messages: face-first contact grid, spoken voice triggers initiate calls. / Landing: PWA install panel, zero manual text inputs to dismiss promos, large install button. NOTE: /messages spec is incomplete in the SRS — requirement clarification session required before this sprint starts (Open Q #5).		
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GATE Phase 2 gate: all 8 routes render · Smart Note-Taker + Health Ledger pass gesture audit and screen reader audit · CI still green

Phase 3 — Polish, Hardening & Release

Timeline: Sprints 5–6 | Weeks 9–12

Goal: WCAG 2.1 AA certified across all 8 routes, full screen reader sign-off (NVDA, TalkBack, VoiceOver), all coverage targets met, all open questions closed, release candidate tagged.

Accessibility — hardening

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 5 Wks 9–10	Full WCAG 2.1 AA audit All routes × all criteria Complete WCAG 2.1 AA audit across all 8 routes: 2.1.1 (keyboard), 2.5.3+4.1.2 (label/speech), 2.5.5 (target sizing), 1.4.3+1.4.4 (contrast + resize). All violations are sprint blockers — no new features until clean.	WCAG 2.1 AA full Deps: All Phase 2 features	Sprint 5
Sprint 6 Wks 11–12	Screen reader sign-off + 200% resize NVDA · TalkBack · VoiceOver; no clipping Manual walkthroughs: NVDA (Windows/desktop), TalkBack (Android), VoiceOver (iOS + macOS). All routes, all interactive elements announced correctly, no focus traps. Separately test all routes at 200% system font scale — no clipping, no layout breaks (rem/em units only).	C4; WCAG 1.4.4 Deps: Full WCAG audit	Sprint 6

QA & performance

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 5 Wks 9–10	Gesture audit — full sweep + coverage All components; 75% RN/Web,	C3 Deps: All Phase 2 features	Sprint 5

	60% Electron/Flutter Systematic gesture audit: no hold >200ms, no drag-and-drop, no infinite scroll, no swipe-required paths. Two-tap loops verified on /medications and /emergency. Simultaneously close coverage gaps: 75% RN/Web, 60% Electron/Flutter.		
Sprint 6 Wks 11–12	Lighthouse final pass + RC ≥90 all routes; release candidate tagged Final Lighthouse audit across all 8 routes on desktop and mobile — all ≥90. Fix regressions from Phase 3 hardening. Produce audit report. Tag release candidate once all gates are green.	CI gate ≥90 Deps: Gesture audit; Coverage	Milestone Do not ship with C1–C4 violations or open WCAG blockers

Open risks — closure

Sprint	Task / Description	Constraints / Dependencies	Risk / Flag
Sprint 5 Wks 9–10	Voice fallback + encryption decision Firefox coverage confirmed; AES-GCM v1 or defer? Validate voice fallback strategy against all target browsers — close Open Q #1. Confirm with stakeholders whether AES-GCM client-side encryption is required for v1 or deferred; if required, implement SubtleCrypto layer over localStorage serialiser — close Open Q #2.	C1 Deps: Voice abstraction spike; State schema	Blocker Voice fallback is a hard C1 requirement — cannot ship without resolution
Sprint 6 Wks 11–12	Final sign-off & release All gates green, docs updated, RC handed off All Phase 3 gates must be green: WCAG certified, screen reader sign-off, Lighthouse ≥90, coverage at target, gesture audit clean, all open questions closed. Produce release notes. Tag RC. Hand off to QA lead for final validation.	— Deps: All Sprint 6 tasks	Milestone

GATE Release gate: WCAG 2.1 AA certified · screen readers signed off · Lighthouse ≥90 · coverage targets met · all open questions closed · RC tagged

Open Architecture Questions & Risks

The following questions must be resolved at the sprint indicated. Unresolved blockers prevent downstream work from starting.

#	Question	Detail	Priority	Action / Timing
#1	Voice fallback strategy	Web Speech API has no Firefox support. Whisper.js WASM fallback must be validated Sprint 1.	High	Sprint 1 — close before scaffold
#2	AES-GCM encryption at rest	Is client-side encryption required for v1 or deferred? Touches entire data layer if added late.	Medium	Clarify with stakeholders Sprint 5
#3	Flutter vs React Native	Benchmark on low-end Android required. Decision gates all mobile scaffold.	High	Sprint 1 — required to unblock scaffold
#4	Multi-device sync	Out of scope per SRS but caregiver coordination may require it. Flag for Phase 2 review.	Medium	Phase 2 planning
#5	/messages spec incomplete	Route requirements table is empty in the SRS. Must clarify before Sprint 4 build starts.	High	Schedule session before Sprint 4